

Experiment 8: Create and Manage a Graph Database

Student Name(s):	Priyanka, Supriya, Amogh, Akshit		
Roll No(s):	40, 39, 36, 38	Experiment Date:	2026-09-19
Quiz Evaluation:	10 / 10 (100%)		

1. Aim & Expected Outcome

Aim: Install Neo4j, create nodes and relationships, and perform basic graph operations.
Expected outcome: A functioning graph database containing connected entities.

2. Learning Objectives

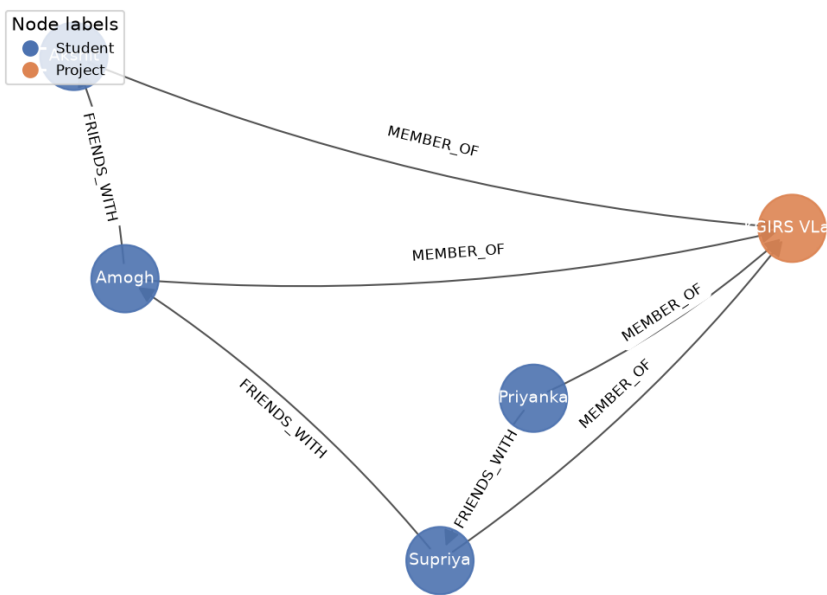
- Understand what a graph database is and how it differs from a relational database.
- Learn the building blocks of a property graph: nodes, labels, relationships and properties.
- Create, read, update and delete (CRUD) nodes and relationships.
- Query the graph for neighbors and shortest paths between entities.
- Visualize the resulting graph and record observations in a lab report.

3. Recorded Experimental Trials & Data

No simulation trials recorded during this session.

4. Final Graph State

Nodes: 5 Relationships: 7



- Priyanka -[FRIENDS_WITH]-> Supriya
- Supriya -[FRIENDS_WITH]-> Amogh
- Amogh -[FRIENDS_WITH]-> Akshit
- Priyanka -[MEMBER_OF]-> KGIRS VLab
- Supriya -[MEMBER_OF]-> KGIRS VLab
- Amogh -[MEMBER_OF]-> KGIRS VLab
- Akshit -[MEMBER_OF]-> KGIRS VLab

5. Observations & Analysis

In this experiment we created a graph database of connected entities using an in-memory simulation of a property graph. We created Student nodes for Priyanka, Supriya, Amogh and Akshit and a Project node called GraphDB Lab, then connected them with directed, typed relationships (FRIENDS_WITH, LEADS and MEMBER_OF). The resulting graph had 5 nodes and 7 relationships, and the graph view showed each node label in a different colour with the relationship types on the edges.

Instructor / Student Signature